

MERRILL, WI, USA

WMO: 726449

Lat: 45.199N

Lon: 89.711W

Elev: 401

StdP: 96.59

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04891

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.3	-23.6	-31.1	0.2	-26.1	-27.9	0.3	-22.6	8.6	-7.5	7.9	-7.1	0.6	290	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	29.8	22.0	27.9	20.9	27.0	20.1	23.6	27.7	22.6	26.4	21.5	25.1	3.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	18.0	25.9	21.3	16.7	24.8	20.3	15.7	23.7	72.5	27.4	68.5	26.1	64.5	25.1	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.4	6.3	-31.7	32.4	1.9	1.8	-33.1	33.7	-34.1	34.7	-35.2	35.7	-36.5	37.0		
			-31.9	25.2	1.8	1.5	-33.2	26.3	-34.3	27.1	-35.3	27.9	-36.6	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-9.9	-9.2	-2.2	5.0	11.8	16.9	19.4	17.9	14.1	6.9	0.0	-6.9
	DBStd	11.62	7.04	6.57	6.97	5.31	4.78	3.57	3.22	3.31	4.51	4.79	5.37	6.22
	HDD10.0	2698	617	537	383	166	35	1	0	1	12	123	300	523
	HDD18.3	4861	876	770	637	400	211	67	26	48	142	355	548	782
	CDD10.0	1018	0	0	4	16	89	208	291	246	135	27	2	0
	CDD18.3	141	0	0	0	1	7	24	59	34	14	2	0	0
	CDH23.3	1594	0	0	1	14	111	279	663	365	151	11	0	0
	CDH26.7	328	0	0	0	2	23	49	162	65	26	1	0	0
(14) Wind	WSAvg	2.3	2.3	2.4	2.5	2.9	2.7	2.2	1.9	1.8	2.0	2.5	2.4	2.3
(15) Precipitation	PrecAvg	830	27	25	48	70	89	97	98	99	112	66	53	37
	PrecMax	1051	85	64	110	111	178	212	239	219	221	130	119	88
	PrecMin	519	1	1	4	21	28	27	38	27	14	5	4	11
	PrecStd	133	19	20	30	26	40	53	47	47	56	33	32	22
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	7.1	21.2	25.9	29.4	31.1	32.5	30.5	29.8	24.9	17.2	8.1
		MCWB	2.9	3.3	15.5	15.9	19.8	23.5	24.6	22.0	22.0	17.1	11.8	5.3
	2%	DB	2.8	3.8	15.0	21.4	26.3	27.7	30.2	28.3	27.2	22.1	13.0	4.8
		MCWB	1.5	1.5	10.1	12.9	18.1	20.7	22.6	21.5	20.2	17.0	9.8	3.5
	5%	DB	1.5	2.1	11.1	17.7	23.8	26.4	28.0	27.1	25.0	18.6	10.9	2.6
		MCWB	0.7	0.4	7.8	10.0	16.0	19.4	21.4	20.2	18.9	14.3	7.6	1.6
	10%	DB	0.0	0.1	7.4	15.1	21.3	24.8	27.1	25.5	22.8	15.6	8.0	1.3
		MCWB	-0.6	-1.0	4.3	8.5	14.6	18.3	20.5	19.4	18.1	11.7	5.8	0.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	4.0	15.7	17.1	21.9	24.5	25.6	24.1	23.2	21.1	13.2	7.2
		MCDB	4.8	6.5	20.9	23.0	27.0	29.3	30.1	27.8	27.2	22.8	15.2	7.5
	2%	WB	1.9	2.1	11.7	14.1	19.6	22.4	23.9	22.9	21.7	17.1	10.7	3.4
		MCDB	2.9	3.8	13.7	20.9	24.0	26.4	28.2	26.2	25.1	20.0	13.3	4.3
	5%	WB	0.6	0.2	7.3	11.7	17.9	21.1	22.9	21.7	20.5	14.8	8.0	1.9
		MCDB	1.5	1.4	10.8	16.0	21.9	24.6	26.8	25.2	23.6	18.2	10.0	2.5
	10%	WB	-0.8	-0.9	4.5	9.2	15.9	19.7	21.7	20.7	19.0	12.3	5.9	0.6
		MCDB	-0.2	0.0	7.3	14.1	19.9	23.3	25.6	23.8	21.9	15.0	8.2	1.2
(35) Mean Daily Temperature Range	MDBR		9.4	11.2	11.6	12.2	13.4	12.6	13.2	13.0	13.2	10.9	8.8	8.2
	5% DB	MCDBR	9.1	11.0	14.7	18.3	16.4	14.3	14.4	14.6	15.2	15.8	13.8	9.2
		MCWBR	7.8	8.8	9.7	10.5	8.5	8.0	7.7	8.0	8.8	10.0	9.9	8.1
	5% WB	MCDBR	8.5	9.8	13.4	16.1	13.3	12.3	12.6	11.9	11.8	13.8	12.1	7.7
		MCWBR	7.4	8.1	9.4	10.4	7.7	7.7	7.5	7.2	7.6	9.8	9.5	7.0
(40) Clear-Sky Solar Irradiance	taub		0.245	0.274	0.313	0.343	0.373	0.372	0.369	0.371	0.345	0.311	0.283	0.253
	taud		2.382	2.298	2.353	2.392	2.346	2.368	2.393	2.391	2.464	2.547	2.529	2.443
	Ebn at Noon		889	914	920	918	895	894	893	878	876	868	838	843
	Edh at Noon		72	96	106	112	121	119	115	111	95	76	64	62
(44) All-Sky Solar Radiation	RadAvg		1.61	2.73	3.89	4.57	5.38	5.86	6.05	5.13	4.03	2.52	1.54	1.14
	RadStd		0.17	0.26	0.40	0.45	0.43	0.47	0.31	0.34	0.28	0.27	0.16	0.13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page